Analysis

PENG

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1 Numbers

1.1 Natural Number system

- This is how mathematics works - it treats its objects abstractly, caring only about what properties the objects have, not what the objects are or what they mean [1].
- It is pointless, at least from a mathematician's standpoint, as to argue about which model is the true one; as long as it obeys all the axioms and does all the right things, that's good enough to do maths.

1.1.1 The Peano Axioms

Axiom 1.1. *Peano axioms:*

1. *0 is a natural number.*
2. *If n is a natural number, then $n{+}{+}$ is also a natural number.*
3. $n{+}{+} \neq 0$
4. $n \neq m \Rightarrow n{+}{+} \neq m{+}{+}$
   $n{+}{+} = m{+}{+} \Rightarrow n = m$
5. *Principle of mathematical induction:*
   Suppose that $P(0)$ is true, and suppose that whenever $P(n)$ is true, $P(n{+}{+})$ is also true. Then $P(n)$ is true for every natural number n.

Definition 1.2.

$$\begin{aligned} 1 &:= 0{+}{+} \\ 2 &:= 1{+}{+} \\ 3 &:= 2{+}{+} \\ &\vdots \end{aligned}$$

1.1.2 Addition

Definition 1.3. *Addition.*

Let n and m be natural numbers.

1. $0 + m := m$
2. $(n{+}{+}) + m := (n + m){+}{+}$

Proposition 1.4. *Let n and m be natural numbers. Then $n + m$ is also a natural number.*

Lemma 1.5. *Let n and m be natural numbers.*

1. $n + 0 = n$
2. $n + (m{+}{+}) = (n + m){+}{+}$

Proposition 1.6. *Let a, b, c be natural numbers.*

1. $a + b = b + a$
2. $(a + b) + c = a + (b + c)$
3. $a + b = a + c \Rightarrow b = c$

Definition 1.7. *A natural number n is positive iff (if and only if) $n \neq 0$.*

Proposition 1.8. *Let a and b be natural numbers.*

1. *If a is positive, then $a + b$ is positive.*
2. $a + b = 0 \Rightarrow a = b = 0.$
3. *If a is positive, then there exists only one natural number b such that $b{+}{+} = a$.*

Proof. Prop. 1.8 (1)

$a + 0 = a$ is positive. Suppose that $a + b$ is positive. Then $a + b{+}{+} = (a + b){+}{+} \neq 0$ is positive. □

Proof. Prop. 1.8 (3)

Suppose that $a = b{+}{+}$. Then $a{+}{+} = (a){+}{+}$. □

Definition 1.9. *Let n and m be natural numbers.*

1. $n \geq m$ *or* $m \leq n$ *iff* $n = m + a$
2. $n > m$ *or* $m < n$ *iff* $n \geq m$ *and* $n \neq m$

Proposition 1.10. *Let a, b, c be natural numbers.*

1. $a \geq a$
2. *If $a \geq b$ and $b \geq c$, then $a \geq c$*
3. *If $a \geq b$ and $b \geq a$, then $a = b$*
4. $a \geq b$ *iff* $a + c \geq b + c$
5. $a < b$ *iff* $a{+}{+} \leq b$
6. $a < b$ *iff* $b = a + d$ *for some positive number d*

Proof. Prop. 1.10 (5)

$b = a + d$ and $b \neq a \Rightarrow d \neq 0 \Rightarrow d = n{+}{+}$

$b = a + (n{+}{+}) = (a{+}{+}) + n \Rightarrow a{+}{+} \leq b$ □

Proposition 1.11. *Let a and b be natural numbers. Then exactly one of following statement is true: $a < b$, $a = b$, or $a > b$.*

Proof. If $a < b$ and $a > b \overset{\text{Prop. 1.10 (3)}}{\Rightarrow} a = b$ contradiction □

Proposition 1.12. *Strong principle of induction.*
Let m_0 be a natural number. Suppose that for each $m \geq m_0$, if $P(m')$ is true for all $m_0 \leq m' < m$, then $P(m)$ is also true. Then $P(m)$ is true for all $m \geq m_0$.

Proof. Let $Q(n)$ be the property that $P(m)$ is true for all $m_0 \leq m < n$.
Suppose that if $Q(n)$ is true, then $P(m)$ is also true.
Then $P(m)$ is true for all $m_0 \leq m < n{+}{+}$. □

Example 1.13. *Fibonacci numbers.*

$$F_n = \frac{\phi^n - \psi^n}{\phi - \psi} \tag{1.1}$$

$$F_{n+2} = F_{n+1} + F_n \tag{1.2}$$

where $\phi = \frac{1+\sqrt{5}}{2}$ *and* $\psi = \frac{1-\sqrt{5}}{2}$

Exercise 1.14. *Principle of backwards induction.*
Let n be a natural number. Whenever $P(m{+}{+})$ is true, $P(m)$ is true. If $P(n)$ is true, then $P(m)$ is true for all $m \leq n$.

Proof. Suppose that if $P(n)$ is true, then $P(m)$ is true for all $m \leq n$. If $P(n{+}{+})$ is true, then $P(m)$ is true for all $m \leq n{+}{+}$. □

Exercise 1.15. *Principle of induction starting from the base case n.*
Let n be a natural number. Whenever $P(m)$ is true, $P(m{+}{+})$ is true. If $P(n)$ is true, then $P(m)$ is true for all $m \geq n$.

Proof. If $P(0)$ is true, suppose that $P(m)$ is true. Then $P(m{+}{+})$ is also true. Then $P(m)$ is true for all $m \geq 0$. Suppose that if $P(n)$ is true, then $P(m)$ is true for all $m \geq n$. If $P(n{+}{+})$ is true, then $P(m)$ is true for all $m \geq n{+}{+}$. □

1.1.3 Multiplication

Definition 1.16. *Let n and m be natural numbers.*

1. $0 \times m := 0$
2. $(n{+}{+}) \times m := (n \times m) + m$

Proposition 1.17. *Let n and m be natural numbers. Then $n \times m$ is also a natural number.*

Lemma 1.18. *Let n and m be natural numbers.*

1. $n \times 0 = 0 \times n$
2. $n \times (m{+}{+}) = (n \times m) + n$

Lemma 1.19. *Commutative.*
Let n and m be natural numbers.

1. $n \times m = m \times n$
2. $n \times m = 0$ *iff at least one of n and m is equal to 0. If n and m are both positive, then nm is also positive.*

Proposition 1.20. *Let a, b, c be natural numbers.*

1. $a(b + c) = (b + c)a = bc + ca$
2. $(ab)c = a(bc)$
3. *If $a < b$ and c is positive, then $ac < nc$.*
4. *If $ac = bc$ and c is positive, then $a = b$.*

Lemma 1.21. *Euclid's division lemma.*

Let n be a natural number and q be a positive natural number. Then there exist natural numbers m and r, such that $0 \leq r < q$ and $n = mq + r$.

Proof. $0 = 0 \times q + 0$ and $0 \leq r < q$. Suppose that $n = mq + r$ and $0 \leq r < q$. Then $n{+}{+} = mq + (r{+}{+})$ and $0 \leq r{+}{+} \leq q$. If $r{+}{+} = q$, $n{+}{+} = mq + q = (m+1)q + 0$. If $r{+}{+} < q$, $n{+}{+} = mq + r{+}{+}$. $\square$

Definition 1.22. *Exponentiation.*

1. $m^0 := 1$ *and* $0^0 := 1$
2. $m^{n++} := m^n \times m$

1.2 Set Theory

References

[1] Terence Tao. *Analysis I*. Vol. 1. Springer, 2022. ISBN: 978-981-19-7261-4.